

# JEE MAIN REVISION MASTER: GRAVITATION

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 14 Analyzed PYQs

## 1. High-Yield Concept Priority Matrix

| Priority Tier | Concept Name                                      | Frequency | Category            |
|---------------|---------------------------------------------------|-----------|---------------------|
| High          | Gravitational Field & Potential                   | 10 PYQs   | Field & Potential   |
| Moderate      | Orbital Velocity, Escape Velocity & Kepler's Laws | 5 PYQs    | Satellites & Orbits |

## 2. Master Formula Reference Card

| Concept / Topic                                   | Mathematical Formulation & Governing Equations                                                                                                           |
|---------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Gravitational Field & Potential                   | $g_h \approx g \left(1 - \frac{2h}{R}\right)$ ; ( $h \ll R$ ), $g_d = g \left(1 - \frac{d}{R}\right)$ , $V(r) = -\frac{GM}{r}$                           |
| Orbital Velocity, Escape Velocity & Kepler's Laws | $v_e = \sqrt{\frac{2GM}{R}} = \sqrt{2gR} \approx 11.2 \text{ km/s}$ , $v_o = \sqrt{\frac{GM}{r}}$ , $v_e = \sqrt{2} v_o$ , $T^2 = \frac{4\pi^2}{GM} r^3$ |

## 3. Deep-Dive Concepts & Linked PYQ Evidence

### 3.1 Gravitational Field & Potential (High Yield — 10 PYQs)

**Core Principle:** Newton's law of gravitation, variation of  $g$  with height, depth, and latitude  $\lambda$  ( $g' = g - \omega^2 R \cos^2 \lambda$ ).

■ **Caution / JEE Pitfall:** At  $h = R$  above Earth,  $g_h = (g / (1 + 1)^2) = (g / 4)$ . The approximation  $(1 - 2h/R)$  fails when  $h$  is comparable to  $R$ !

■ **Speed Trick:** Decrease in  $g$  at height  $h$  is double the decrease at depth  $d$  for  $h = d \ll R$ .

**Representative Source PYQs:**

- [2026 Shift] Q#1 (MCQ): "Suppose the acceleration due to gravity at the earth's surface is  $gm/s^2$  and at the surface of moon it is  $g^m/s^2$  ...." [Ans: N/A]
- [2026 Shift] Q#2 (MCQ): "Imagine a new planet having the same density as that of the earth, but it is two times bigger than the earth in size. If the acceleration du..." [Ans: N/A]
- [2025 Shift] Q#4 (MCQ): "The total energy of a satellite in a circular orbit at a distance  $(R + \blacksquare)$  from the centre of the Earth varies as  $[R \text{ is the radius of the ...}]$ " [Ans: N/A]

### 3.2 Orbital Velocity, Escape Velocity & Kepler's Laws (Moderate Yield — 5 PYQs)

**Core Principle:** Orbital mechanics of satellites, escape speed from planetary surfaces, and Kepler's three laws of planetary motion.

■ **Caution / JEE Pitfall:** Total energy of a bound satellite is negative:  $E = -K = (1/2)U = -(GMm/2r)$ .

■ **Speed Trick:** Geostationary satellite period is 24 hours, orbits at height  $\approx 36,000$  km strictly in the equatorial plane from West to East.

**Representative Source PYQs:**

- **[2025 Shift] Q#3 (MCQ):** "If  $r_p, v_p, L_p$  and  $r_a, v_a, L_a$  are radii, velocities and angular momenta of a planet at perihelion and aphelion of its ellip..." **[Ans: N/A]**
- **[2025 Shift] Q#4 (MCQ):** "The total energy of a satellite in a circular orbit at a distance  $(R + \blacksquare)$  from the centre of the Earth varies as  $[R]$  is the radius of the ..." **[Ans: N/A]**
- **[2023 Shift] Q#2 (MCQ):** "When a planet revolves around the Sun, in general, for the planet..." **[Ans: N/A]**

## 4. Benchmark Illustrative PYQs with Master Solutions

### Benchmark Problem 1: KCET 2026 (Q#1)

Suppose the acceleration due to gravity at the earth's surface is  $g\text{m/s}^2$  and at the surface of moon it is  $g^{\text{m}}\text{m/s}^2$ . An  $M$  kg passenger goes from the earth to the moon in a spaceship moving with a constant velocity (Neglect all other objects in the sky). Which curve best represents the weight (net gravitational force) as a function of time?

- (A)
- (B)
- (C)
- (D)

**Correct Answer: None**

**Solution Analysis:** Let the mass of the Earth be  $M_e$ , the mass of the Moon be  $M_m$ , and the distance between their centres be  $d$ . Suppose the spaceship is at a distance  $r$  from the centre of the Earth. Then its distance from the centre of the Moon is  $(d-r)$ . The Earth pulls the passenger towards the Earth, and the Moon pulls the passenger towards the Moon. These two pulls are in opposite directions, so the net gravitational force on the passenger of mass  $M$  is:  
 $F_{\text{net}} = \frac{GM_e M}{r^2} - \frac{GM_m M}{(d-r)^2}$  The spaceship moves with constant velocity. So the distance  $r$  from the Earth in

**Benchmark Problem 2: KCET 2025 (Q#3)**

If  $r_p, v_p, L_p$  and  $r_a, v_a, L_a$  are radii, velocities and angular momenta of a planet at perihelion and aphelion of its elliptical orbit around the Sun respectively, then

- (A)  $r_p > r_a, v_p > v_a, L_a > L_p$
- (B)  $r_p < r_a, v_p > v_a, L_a = L_p$
- (C)  $r_p > r_a, v_p < v_a, L_a = L_p$
- (D)  $r_p < r_a, v_p < v_a, L_a < L_p$

**Correct Answer: None**

**Solution Analysis:** Angular momentum is conserved in an elliptical orbit, meaning:  $L_p = L_a$  where  $L_p$  is the angular momentum at perihelion and  $L_a$  is the angular momentum at aphelion. The perihelion is the point in the orbit where the planet is closest to the Sun. Therefore, the radius at perihelion ( $r_p$ ) is less than the radius at aphelion ( $r_a$ ):

$r_p < r_a$  Because angular momentum is conserved, we have:  $mv_p r_p = mv_a r_a$  From this, it follows that the velocity at perihelion ( $v_p$ ) is greater than the velocity at aphelion ( $v_a$ ):  $v_p > v_a$